CLINICOS: DESIGNING DOCUMENTATION FOR THE TWO-MINUTE CONSULTATION

A Design Thinking Report on a Consultation-Documentation Assistant for Single-Doctor Rural Clinics in India

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Topic

ClinicOS – a local-first assistant that records, transcribes and drafts the clinical note, so the doctor only reviews and signs

Subject

Design Thinking

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INTRODUCTION

ClinicOS is a consultation-documentation assistant built for single-doctor clinics in rural India, with its field site in Krishnanagar, Nadia district, West Bengal. In such a clinic the doctor has roughly two minutes for each patient, and in that time must greet, take a history, examine, explain, prescribe and record. The record is the part that gets dropped.

ClinicOS does not ask the doctor to type. It records the consultation that was already happening, transcribes it on the clinic's own computer, drafts a structured clinical note, and asks the doctor only to read, correct and sign. A branded prescription prints from the signed note. The doctor stays the only person who makes clinical decisions; the software carries the record, it does not judge it.

This report follows the five stages of design thinking shown in our course material — **Empathize, Define, Ideate, Prototype and test, and Implement** — and explains what each stage produced for ClinicOS. Section 6 contains screenshots of the working prototype.

Stage	Question it answers	What it produced for ClinicOS
1. Empathize	Who are we designing for, and what is their day really like?	Research on consultation time and documentation load; a discovery interview script; a persona, an empathy map and a journey map
2. Define	What exactly is the problem?	Four problem clusters, one problem statement, a point of view and six How-Might-We questions
3. Ideate	What could solve it?	A backlog of 37 ideas, four directions compared, one chosen, and four design principles
4. Prototype and test	Does the idea work, and how will we know?	Working software, automated test suites, speed measurements and pass/fail gates set in advance
5. Implement	How does the prototype become a product?	Roles, front desk, billing, follow-up, a patient portal and an online service, with safeguards enforced in code

1. EMPATHIZE

Conduct research to understand users' problems and needs.

1.1 The Setting: One Room, One Doctor, One Register

India's rural primary care runs on very small units. As of 31 March 2023 the country had 31,882 Primary Health Centres staffed by 40,583 doctors — roughly **1.3 doctors per centre** (Health Dynamics of India 2022–23, Ministry of Health and Family Welfare). Alongside this public network sits a much larger and largely uncounted layer of private single-doctor chambers: one room, one doctor, a paper register and a queue outside.

That chamber is the unit this project studies. It is where the largest number of Indian patients are seen, and almost no clinical software is designed for it. In a single-doctor clinic the doctor is at once the receptionist, the diagnostician, the writer of instructions for the pharmacist and the record-keeper.

1.2 How We Set Out to Listen

We chose methods that reveal what doctors actually do rather than what they say they would do:

- **Secondary research.** Published studies on consultation length, physician time use and rural internet access, used to understand the shape of the problem before going into the field.
- **Discovery interviews with 20 doctors.** A 20-minute script that never mentions software in its first half and asks only about the past — what the doctor did yesterday — because a doctor asked "would you use an AI scribe?" will politely say yes, while a doctor asked "what took the longest yesterday?" will tell the truth.
- **Observation in the room.** Watching consultations end to end and timing what really happens, rather than asking a doctor to recall it afterwards.
- **Co-designing the consent line.** Writing, in Bengali and English, the sentence a doctor will say aloud to a patient before recording, together with the people who will have to say it in front of a queue.

The interview script follows five rules: do not open the laptop for the first fifteen minutes; never ask "would you use this?" or "would you pay for this?"; ask about the past, not the future; write notes after the conversation, not during it; and ask for two further introductions at the end of every interview.

Honest status of this stage. At the time of writing the secondary research is complete and the interview script is ready, but the twenty discovery interviews and the observation sessions have not yet been carried out. They are the first task of the next phase (Section 5.4). Every finding in Section 1.3 is therefore from published evidence and is labelled with its source; the persona and empathy map in Sections 1.6 and 1.7 are hypotheses to be checked against the interviews, not interview results.

1.3 What the Evidence Says

Finding	Figure	Source
Average primary-care consultation in India	about 2 minutes (22.5 minutes in Sweden)	Irving et al., BMJ Open, 2017 (67 countries)
Countries where patients get 5 minutes or less	18 countries, about half the world's population	Irving et al., 2017
Physician office time spent on records and desk work	49.2%, against 27.0% with the patient	Sinsky et al., Annals of Internal Medicine, 2016
Documentation after clinic hours	a further 1–2 hours a day	Sinsky et al., 2016
Rural households with internet access	83.3% (urban 91.6%)	NSS 80th Round, Telecom, Jan–Mar 2025

Two minutes must contain the greeting, the history, the examination, the explanation, the prescription and the record. What survives in practice is a line in a register or a carbon copy of the prescription — enough to hand the patient, but not enough to reconstruct the visit six months later.

1.4 What We Still Have to Hear Ourselves

Published evidence gives the shape of the problem; it cannot give one clinic's version of it. These are the questions the interviews are designed to answer:

1. What actually took the longest in your clinic yesterday?
2. When a patient you saw six months ago walks back in, how do you find out what you gave them last time?
3. What happens to the register when it is full?
4. Have you ever used software in the clinic? If you stopped, what made you stop?
5. Of the patients you ask to come back, how many do?
6. What would make you stop using something on day three?

1.5 The Assumptions That Broke

Two of our starting assumptions did not survive the research, and changing them changed the design:

We assumed	The evidence said	So we reframed
Rural clinics are offline; connectivity is the binding constraint.	83.3% of rural households have internet access, against 91.6% urban — a real gap, but far smaller than we assumed.	The constraint is not the absence of a connection but what a dropped one costs mid-consultation. The target became "never loses a consultation, for any reason."
Give the clinic an electronic record and the paperwork problem goes away.	Physicians spend 49.2% of office time on records and desk work against 27.0% with the patient.	Software that asks a doctor to type during a two-minute visit does not return time; it takes it. The remedy must cost less attention than paper.

1.6 Who We Are Designing For: Persona

The persona below is a composite built from the published evidence and the field setting. Dr. Roy is not a real individual; the interviews will confirm or correct him.

Persona	Dr. A. Roy, 46 — solo practitioner, rural West Bengal
Practice	40–60 patients on a working day; no receptionist or assistant; one laptop and one paper register
Languages	Speaks Bengali with patients; writes prescriptions in English
What he needs	To finish each visit with a usable written record without spending the visit producing it
What frustrates him	Recalling a patient he last saw eight months ago from one line in a register — or not recalling them at all
What he will not tolerate	Anything that adds a step in front of a waiting patient, or that loses a consultation even once

1.7 Empathy Map (Hypotheses)

Says	Thinks
"I don't have time to write everything." "The patient never brings the old prescription." "Computers slow me down."	Every second spent writing is taken from the patient. If the laptop dies, years of records go with it. Software is made for big hospitals, not for a clinic like mine.
Does	**Feels**
Talks, examines and prescribes in about two minutes. Writes only the prescription; the rest stays in his head. Relies on the patient to remember earlier visits.	Rushed by the queue outside. Uneasy when a returning patient's history is unknown. Wary of any tool that has failed him before.

1.8 Journey Map: The Consultation as It Happens Today

Step	What happens	What is recorded	Pain point
1. Patient arrives	Name called from the queue	Sometimes a line in the register	No history in front of the doctor
2. History taken	Patient describes the complaint in Bengali	Nothing	Details exist only in the doctor's memory
3. Examination	Doctor examines and notes findings mentally	Nothing	Findings are lost after the visit
4. Explanation	Doctor explains the condition and the plan	Nothing	Patient forgets instructions
5. Prescription	Doctor writes medicines by hand	The prescription, for the pharmacist	Often illegible; not written for the next visit
6. Patient leaves	Next patient called	—	The next visit starts from nothing

The breakdown point. Everything the doctor learns between steps 2 and 4 exists only in his head and in the two minutes he has. By step 6 it is gone, and only the prescription reliably becomes a record — and it is written for the pharmacist, not for the next visit.

2. DEFINE

Analyze insights to define core user problems.

2.1 Four Problems, Not One

Clustering what we found produced four distinct problems. Solving only the first is how most clinical software fails.

#	Problem	Why it matters
1	Time is the scarcest resource	Two minutes must hold the greeting, the history, the examination, the explanation and the record.
2	The record is sacrificed first	It is the only part of the visit with no one in the room speaking up for it.
3	Continuity dies with the register	A patient's history is only as durable as one notebook and one laptop.
4	Consent is unwritten	Nobody in this workflow has ever been asked to record permission, and data-protection law now requires it.

2.2 Problem Statement

> *In single-doctor rural clinics, a consultation lasting roughly two minutes must produce both care and a record, and the record is routinely the part that is sacrificed. Existing clinical software addresses this by asking the doctor to type structured data during the consultation, which measurably consumes more of the doctor's attention than paper does and is consequently abandoned. Clinics are left with no usable patient history, unsystematic follow-up, and years of records resting on a single register or a single laptop — while any acceptable remedy must also avoid clinical judgements it is not permitted to make, and must protect consultation content that is legally sensitive personal data.*

2.3 Point of View

> *Dr. Roy, a solo practitioner seeing 40–60 patients a day, needs a way to end each consultation with a usable written record without spending consultation time producing it — because the two minutes he has belong to the patient, and every second the record takes is taken from care.*

2.4 How Might We…

1. How might we let a record write itself from what is already being said in the room?
2. How might we make recording feel like less work than the paper it replaces?
3. How might we guarantee that no consultation is ever lost, whatever fails — the browser tab, the laptop, the power or the network?
4. How might we ask a patient for meaningful consent in under fifteen seconds, in their own language?

5. How might we keep a clinic's history portable enough that the clinic never feels captured by us?
6. How might we stop short of clinical judgement while still being genuinely useful?

2.5 The Boundaries We Designed Inside

Two legal boundaries shape the solution rather than merely decorating it:

- **CDSCO — Software as a Medical Device.** Software limited to storing, formatting and communicating data sits outside medical-device regulation in India. Software that quantifies a physiological parameter to drive a clinical decision does not. A system that generated diagnoses, risk scores or triage badges would change what the project legally is.
- **Digital Personal Data Protection Act, 2023 (and the DPDP Rules, 2025).** Consultation content is personal data of the most sensitive kind. Informed consent, purpose limitation and deletion rights apply to a student prototype exactly as they apply to an established company.

Both point in the same direction: the system should **carry** the record, not **judge** it.

2.6 Success Criteria, Set Before Testing

Criterion	Measure	Threshold
The problem is real	Doctors naming documentation, unprompted, in their top three frustrations	at least 6 of 20
It is adopted, not endured	Pilot clinics still using it unattended on day 21	at least 2 of 3
It returns attention	Doctor-reported time per consultation against the baseline	not worse
The record is trustworthy	Doctor edits needed per AI-drafted note	trending down
Nothing is ever lost	Consultations lost to a crash, power cut or network failure	0

3. IDEATE

Brainstorm innovative solutions to the core problems.

3.1 Brainstorming: A Wide Field of Ideas

Brainstorming against the six How-Might-We questions produced 37 ideas, which we grouped into tiers by how much each depends on the others and how much risk it carries. A selection:

Tier	Example ideas	Why it matters
0 — Foundation	Crash and offline recovery; one-tap backup and restore; a speed pass; an activity log	A doctor who loses one consultation in front of a waiting patient stops using the system that afternoon
1 — Daily use	A token queue for the front desk; search by phone number; a Bengali medicine-timing card; WhatsApp follow-up reminders	Keeps the clinic using it even on days the doctor is too busy to record
2 — Value to the owner	Billing and receipts; a day-end summary; the follow-up list valued at the clinic's own fee	Gives the clinic owner a reason to pay that they can verify themselves
3 — Speed in the room	The doctor's own favourite prescriptions; repeat last visit's medicines; a quick visit without recording; search from any screen	Saves seconds on every one of 40–60 visits
4 — Records depth	Household linking by phone; the last visit at a glance; the clinic's logo on the prescription	Makes the second visit better than the first
5 — Chronic care (risky)	Trends of readings; doctor-set flags	Marked risky: anything computed into a risk badge crosses into regulated medical-device territory

3.2 Four Directions Compared

Direction	What it is	Verdict
Faster paper	Printed templates and tick-boxes	Rejected: still costs attention during the visit, and produces a record nobody can search
A conventional clinic EMR	Typed, structured fields on screen	Rejected: the measured failure mode — documentation time goes up and the software is abandoned within weeks
A cloud AI scribe	Audio uploaded to a remote service for transcription and drafting	Rejected as the default: sends the most sensitive audio a clinic produces off the premises, and stops when the link drops
An on-device ambient scribe	Listens to the consultation already happening, drafts the note locally, asks the doctor only to review and sign	**Chosen**

Three of the four directions were rejected on evidence gathered in the Empathize stage, not on preference.

3.3 Selection Criteria

Criterion	Faster paper	Clinic EMR	Cloud AI scribe	On-device scribe
No typing during the visit	No	No	Yes	Yes
Works when the internet drops	Yes	Partly	No	Yes
Audio stays where the clinic chose	Yes	Yes	No	Yes
Searchable history	No	Yes	Yes	Yes
Runs on an ordinary laptop at low cost	Yes	Yes	Partly	Yes

3.4 Four Design Principles

Each principle answers one How-Might-We question, and each has cost us a feature we might otherwise have built:

- **Listen, don't ask.** The record is drafted from the conversation that was going to happen anyway. The doctor is never asked to enter data mid-visit.
- **Never lose a consultation.** Audio is written to disk continuously, not at the end. Work in progress survives a crash, a closed laptop or a power cut.

- **Carry, don't judge.** The system transcribes, structures and prints. It does not diagnose, score or advise — that boundary is deliberate.
- **Leave the door open.** Every record can be exported in an open format at any moment. A clinic that wants to leave should find it easy.

3.5 What We Chose Not to Build

Not built	Why
Diagnosis suggestions, risk scores, "high attention" flags	Clinical judgement belongs to the doctor, and it would make ClinicOS a regulated medical device
Keeping the audio after signing	Patients are promised the recording is deleted the moment the doctor signs
Training AI models on patient audio	A settled decision; only de-identified data, with an opt-out, may ever be used
Pre-written treatment templates	A shipped "viral fever" template would be clinical advice; each doctor saves only their own favourites
Lock-in	A full export in an open format must always work, even for a paused clinic

4. PROTOTYPE AND TEST

Build prototypes, test often, and refine solutions based on feedback.

4.1 What Was Built: Working Software, Not a Mock-up

The prototype is a working application that runs on an ordinary clinic laptop with no internet connection. A consultation moves through four steps:

Step	What the doctor does	What ClinicOS does
1. Record	Presses Start recording and talks to the patient as usual, in Bengali, Hindi, English or a mix	Saves the audio to disk in 5-second pieces, so an interruption loses seconds, not the visit
2. Transcribe	Nothing — transcription starts by itself when recording stops	Converts speech to text on the clinic's own computer (faster-whisper)
3. Draft	Nothing	Drafts a structured note — complaint, history, examination, diagnosis, medicines, tests, advice — using a local language model (Qwen 2.5, 7B, through Ollama)
4. Review and sign	Reads the draft, corrects anything wrong, adds a follow-up date or warning signs, and signs	Files the note, deletes the recording, logs every correction, and prints a branded prescription

If the patient says no to recording, or the visit is trivial, the doctor writes the note by hand on the same screen; nothing is recorded.

4.2 How It Is Built

- **Application:** Next.js 15, React 19 and TypeScript, with a hand-written design system and plain-language screens.
- **Data:** SQLite on the clinic's own computer; PostgreSQL for the online version. Hand-written SQL, no ORM.
- **AI behind small interfaces:** speech-to-text and note drafting each sit behind a provider interface, so a hosted service can replace the local one through settings alone.
- **Recovery:** the note autosaves to the server and to the browser's own storage; unfinished consultations reappear on the Today screen with the reason they stopped.
- **Backup:** a daily automatic snapshot (14 kept) taken with SQLite's online backup API — never a plain file copy, which can silently miss recent consultations — and a restore guarded by typing RESTORE.

4.3 Design Decisions Traced Back to the Field

What we learned	What we built
"A consultation must survive anything."	Audio streamed to disk every 5 seconds; the note saved in two independent places; interrupted visits shown with the words "nothing here has been lost".
"Consent has to fit in the room."	A short spoken explanation in Bengali and English on the recorder screen, and a server that refuses to record until the clinic has accepted ClinicOS's terms.
"One laptop is one point of failure."	A verified daily snapshot of the whole clinic, restorable in one step, and a plain-file export that works without ClinicOS at all.
"The doctor must stay in charge."	The AI draft is stored unchanged; the doctor's edits are stored separately; the difference at signing is the accuracy measure; signing is labelled as the clinical act.
"Doctors disbelieve inflated numbers."	Time saved is counted at a deliberately conservative 4 minutes per AI-drafted note, and never for notes written by hand.

4.4 How the Prototype Was Tested

Automated end-to-end tests. The test suites build the real application against a throwaway database, start it, and exercise it over real HTTP — every page and every route, with each role, and with another clinic's record ids to prove that one clinic can never reach another's data. On 20 September 2026 all **539 checks passed on PostgreSQL and 357 on SQLite**.

Test suite	Checks
Upgrade safety and the permission matrix	144
The day's list and search by phone	50
The doctor's speed features	55
Prescription extras	18
Billing and follow-ups	70
Password reset and restoring an old backup	14
Many clinics in one database (online only)	73
Clinic computer sync (online only)	71
The patient portal (online only)	44

Speed as a clinic grows. A benchmark builds a throwaway clinic of 5,000 patients and about 10,000 visits and times each screen. Measured on a Windows laptop: Today 154 ms, Patients 50 ms, a patient's record 32 ms, Bills 33 ms, Follow-ups 79 ms, Clinic admin 225 ms, and exporting the whole clinic 800 ms for 35 MB.

Testing with users. The tests that decide whether ClinicOS continues are the gates in Section 2.6, which were set before any testing so that they cannot be bent to fit the results. Failing any gate stops the project at that gate. To make the field test measurable, the prototype itself records the edits the doctor makes to each AI draft and how long each step of a consultation takes (Figure 17).

4.5 Refining the Solution from Feedback

Reviewing each working build inside the team led to these refinements. Feedback from doctors will come through the interviews and pilots.

- **No consent screen on every visit.** A doctor will not tap through a screen for each patient. The clinic now accepts ClinicOS's terms once — undertaking to tell every patient and to record no one who says no — and the server still refuses to record until it has.
- **A path for "no".** A patient who declines recording is not sent back to paper: the doctor writes the note by hand on the same screen.
- **Safety fixes.** An empty note can no longer be signed, and a signed visit refuses to be transcribed or drafted again, which could have overwritten the draft it was signed against.
- **Honest measures.** Step times are shown as the middle value and the longest, never an average, because one visit left open over lunch would distort an average.
- **Lighter software.** Six animation libraries were removed from the clinic app (177 MB less on every clinic computer), and the public website was rebuilt as a light page instead of a heavy scroll-driven film.

5. IMPLEMENT

Turn prototypes into functioning products.

5.1 From Prototype to Product

The single-doctor prototype was extended into a product that a whole clinic can run on. Each addition traces back to a backlog idea from the Ideate stage:

- **Accounts, roles and an activity log.** Doctor, front desk and clinic admin, each checked by the server on every page and request — hiding a button is never the control.
- **A front desk that runs the day.** Find a patient by name or phone, register them onto today's numbered list, and see who is waiting; the doctor sees "Waiting now" with Start visit.
- **Speed features for the doctor.** Repeat the last visit's medicines in one tap, the doctor's own favourite prescriptions, a quick visit without recording, and Ctrl+K search from any screen.
- **Prescription extras.** The clinic's logo, a "come back at once if…" box in the doctor's own words, and a medicine-timing card in Bengali, Hindi or English that only re-reads what the doctor wrote.
- **Billing and follow-up.** Bills, payment by cash, UPI or card, printed receipts with nothing clinical on them, and a follow-up list with a WhatsApp reminder that a person sends — ClinicOS sends nothing itself.
- **A patient portal.** After signing, the doctor can share the prescription with the patient, who opens it on their phone in English, Bengali or Hindi with a printed code and a PIN.
- **An online service.** Many clinics in one database, each seeing only its own records, with an optional paid add-on that keeps the clinic's own computer in step with its online copy.

Capability	Doctor	Front desk	Clinic admin
Clinical: transcripts, notes, signing, prescriptions	Yes	No	No
Front desk: register, find patients, today's list, follow-ups	Yes	Yes	No
Admin: people, clinic details, backups, export, deletion	If owner	No	Yes
Billing: bills, payments, receipts	Yes	Yes	Yes

5.2 Where It Runs

One codebase runs two ways. **On the clinic's own computer** it uses SQLite, needs no internet and sends nothing anywhere. **Online**, at vxclinicos.vercel.app/app, it runs on Vercel with its database in Supabase's Mumbai region. The online copy is still a prototype: it says so on every screen and is for made-up patients only, until the patient consent script covers online processing. A public website describes the product and takes enquiries.

5.3 Safeguards Enforced in Code

1. No clinical judgement: no diagnosis generation, risk scores or triage badges.
2. Identifiable data goes only where the clinic chose to keep it; one clinic's records are never shown to another, and Varaxis's own console shows counts only.
3. Audio is deleted when the doctor signs.
4. Patient audio is never used to train AI.
5. Recording is allowed only on a basis the server checks.
6. The AI's original draft is never overwritten, so its accuracy can be measured honestly.
7. Backups use SQLite's online backup API, never a file copy.
8. No lock-in: a full export in an open format always works.

5.4 Next Steps

When	Step	Gate
Weeks 1–4	Carry out the 20 discovery interviews with the script in Section 1.2	At least 6 of 20 doctors name documentation unprompted
Weeks 5–8	Pilot in 3 clinics, unattended after setup	At least 2 of 3 still using it on day 21
Weeks 9–16	Convert pilots to paying clinics; finish the consent script and agreements for online use	At least 5 clinics paying
Throughout	Watch for any lost consultation	Zero, always

Building further features before these interviews is the project's main risk. The next step is to listen.

6. PROTOTYPE SCREENSHOTS

The screenshots below were taken from a sandbox copy of the working prototype on 12 October 2026. The sandbox is filled with **made-up patients**; no real patient's data appears. The sandbox was not running a speech model, so the transcript and AI draft in Figures 6 and 7 are sample text entered for illustration; the screens around them are exactly as the software shows them.

6.1 Getting In

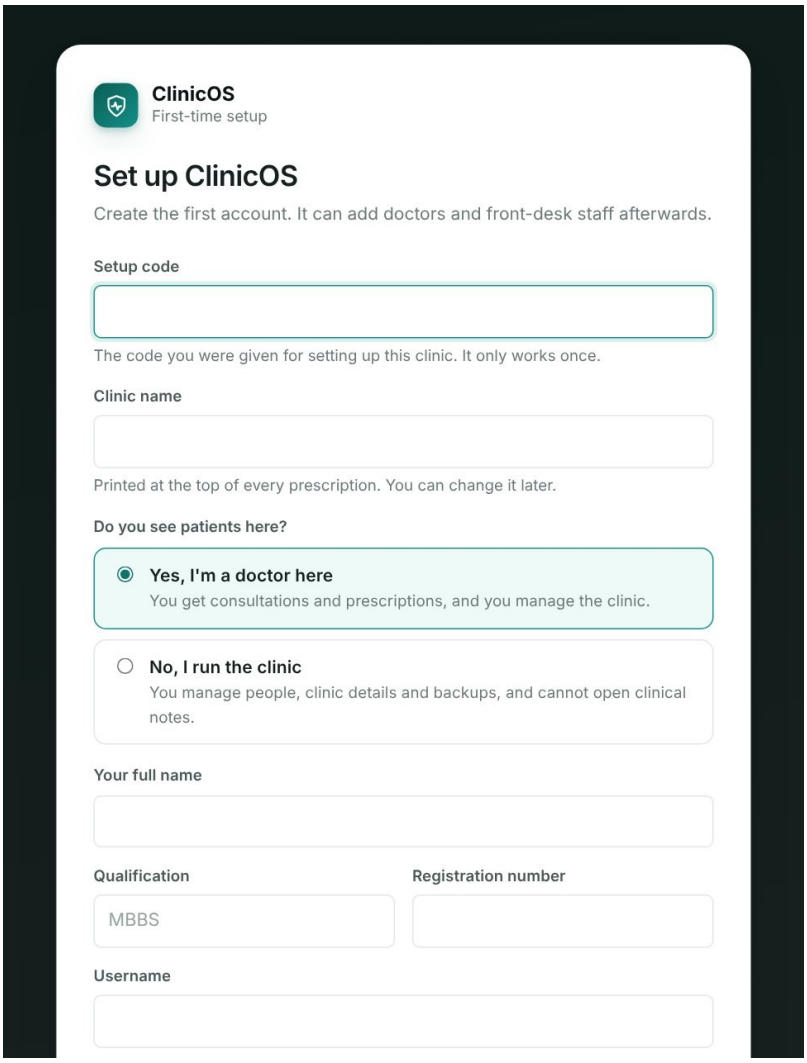


Figure 1: *First-run setup. The first account is created with a one-time setup code, and setup then refuses forever.*

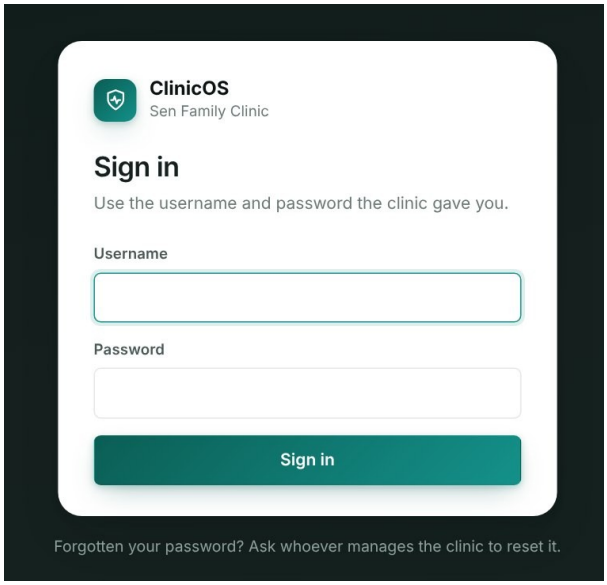


Figure 2: *Sign-in. Each person has their own account; there is no shared clinic password.*

6.2 The Doctor's Consultation

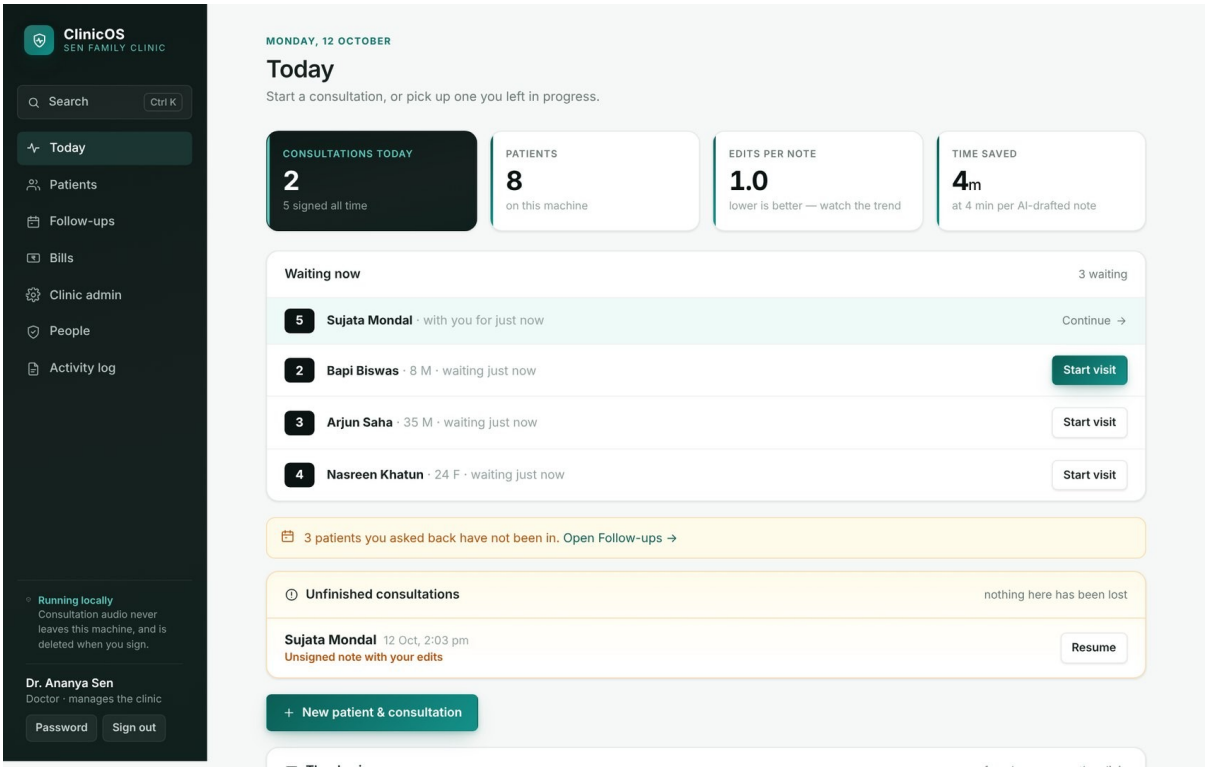


Figure 3: *The doctor's Today screen: honest numbers at the top (including edits per note), patients waiting, a follow-up reminder, and unfinished consultations marked "nothing here has been lost".*

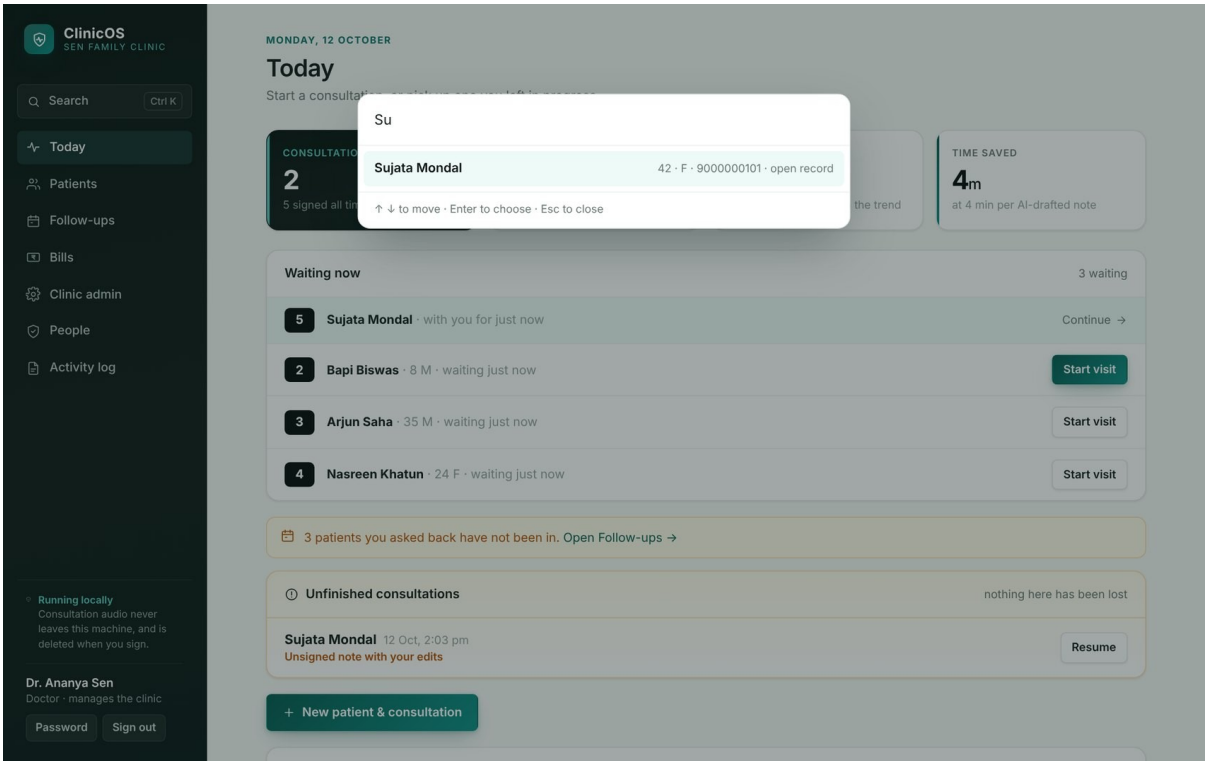


Figure 4: *Ctrl+K search from any screen finds a patient by name or phone number.*

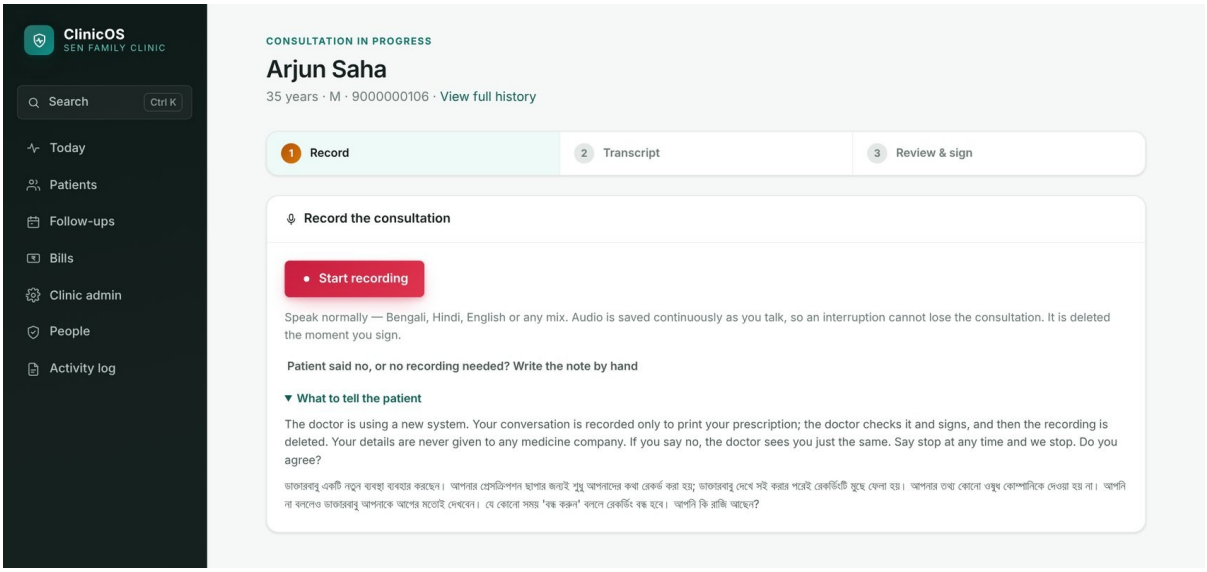


Figure 5: *Step 1, Record. The spoken explanation for the patient is shown in English and Bengali; a patient who says no gets a note written by hand instead.*

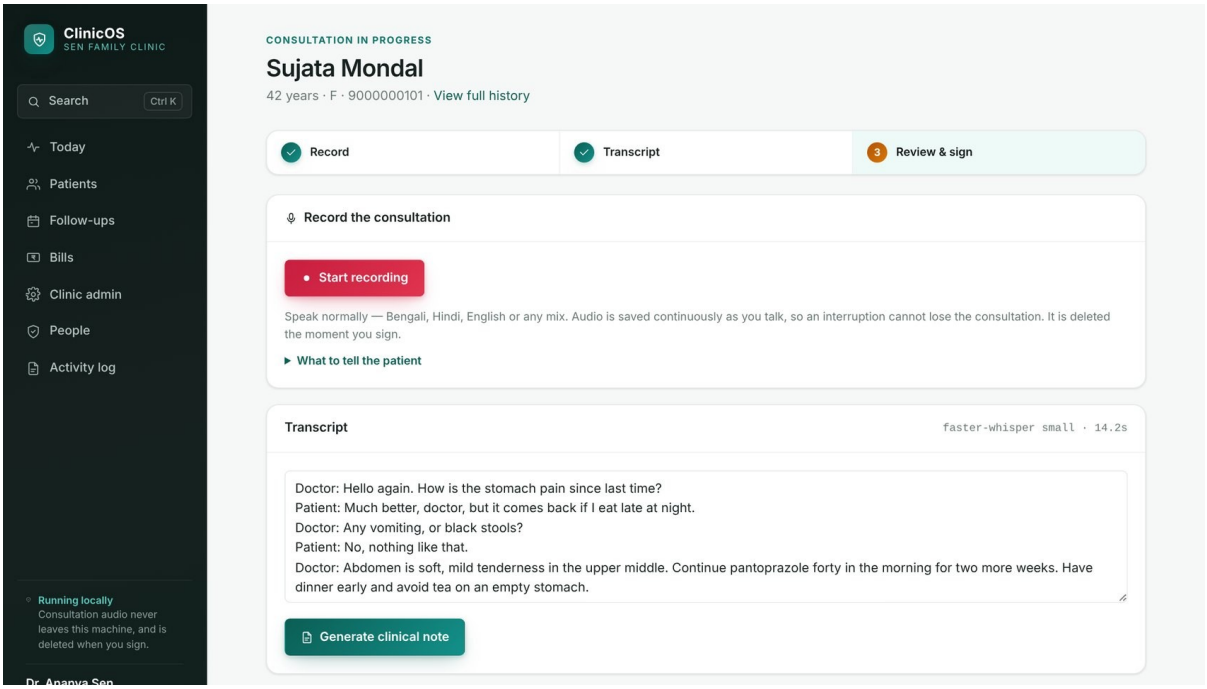


Figure 6: *Steps 1–3 in progress: the recording, its transcript (sample text), and the button that drafts the clinical note.*

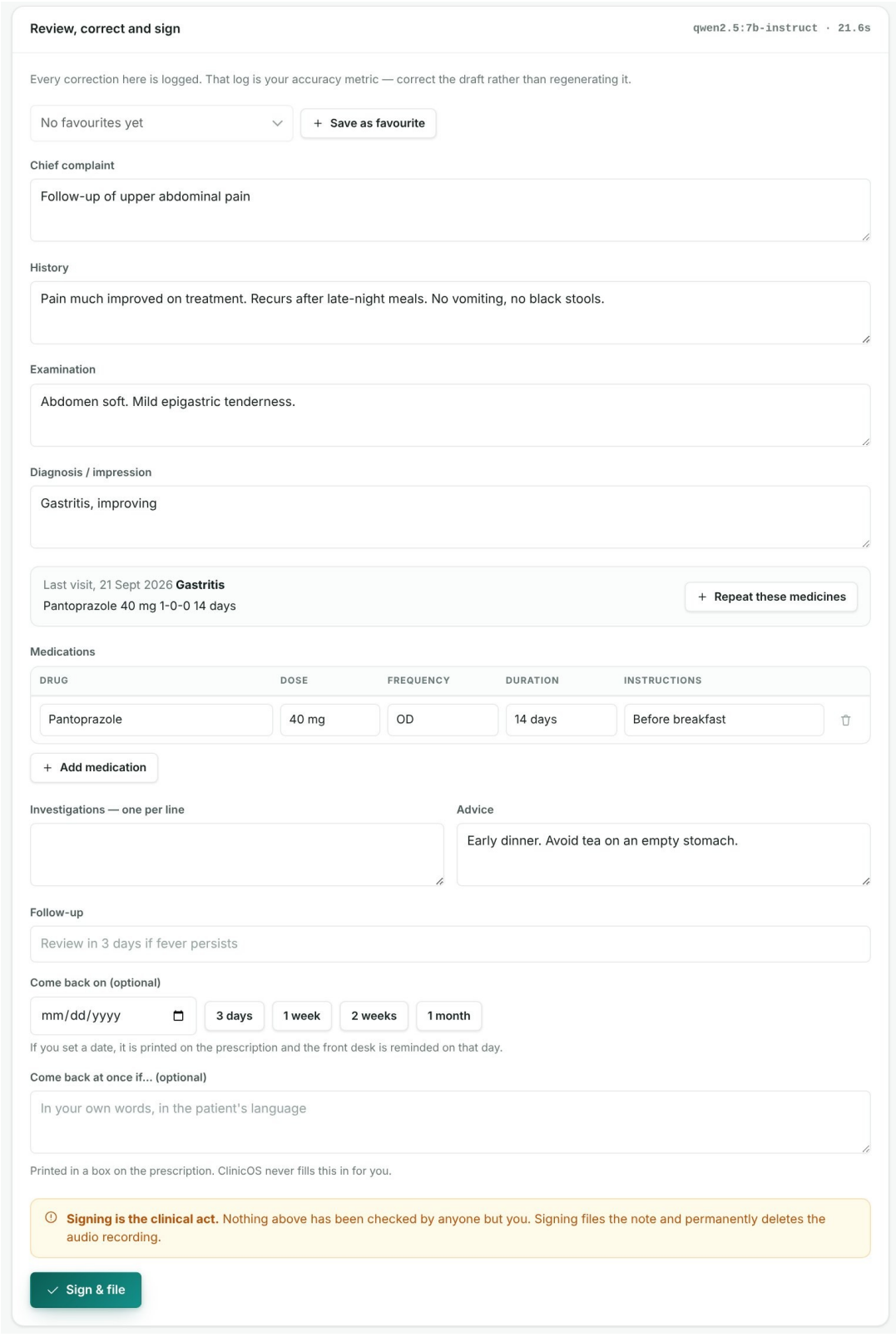


Figure 7: *Review, correct and sign. The AI draft is editable; the patient's last visit is shown with one-tap repeat of its medicines; the follow-up date and warning signs are written only by the doctor; signing is labelled as the clinical act.*

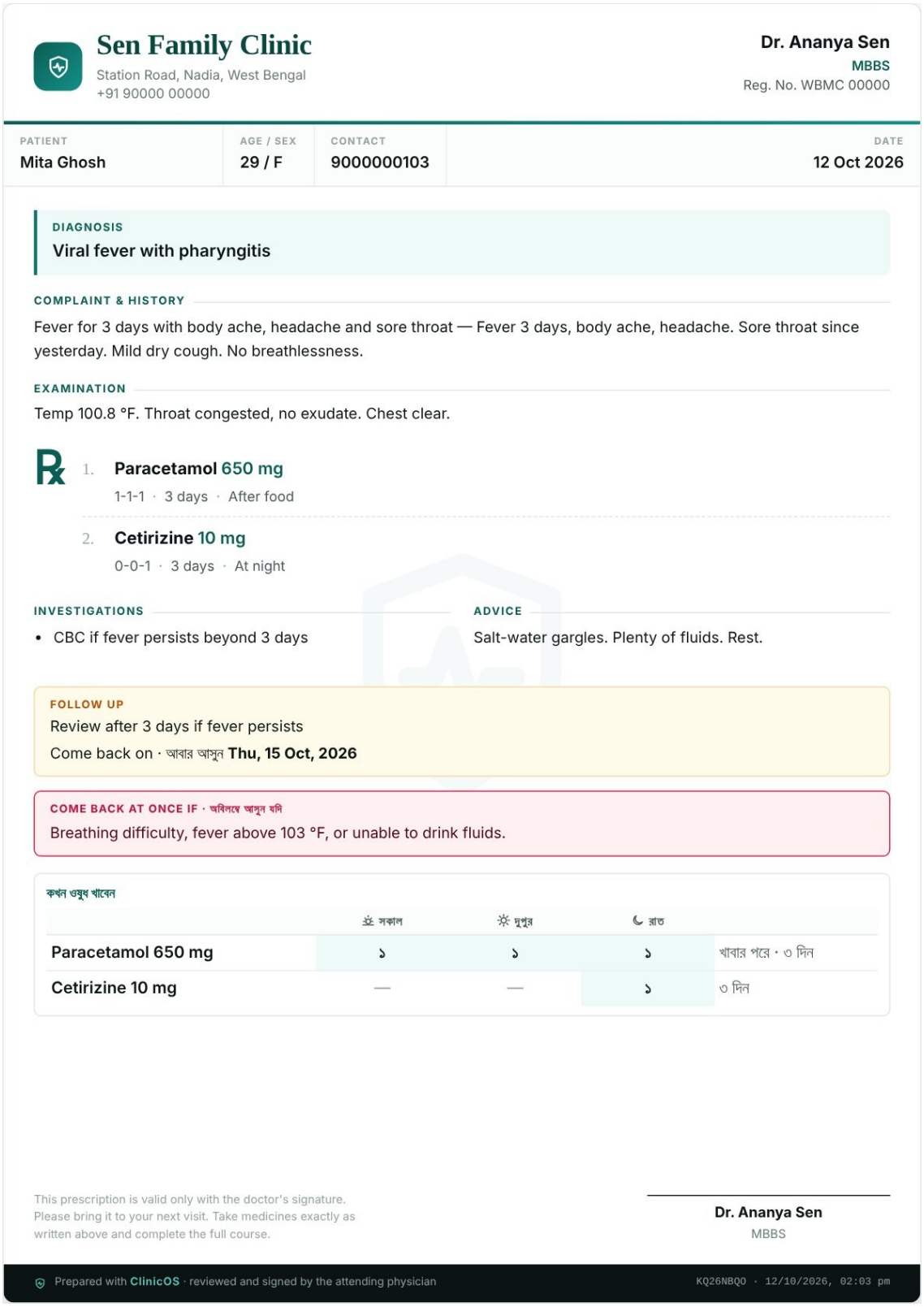


Figure 8: *The printed prescription generated from the signed note, with the come-back date, the "come back at once if…" box and the medicine-timing card in Bengali.*

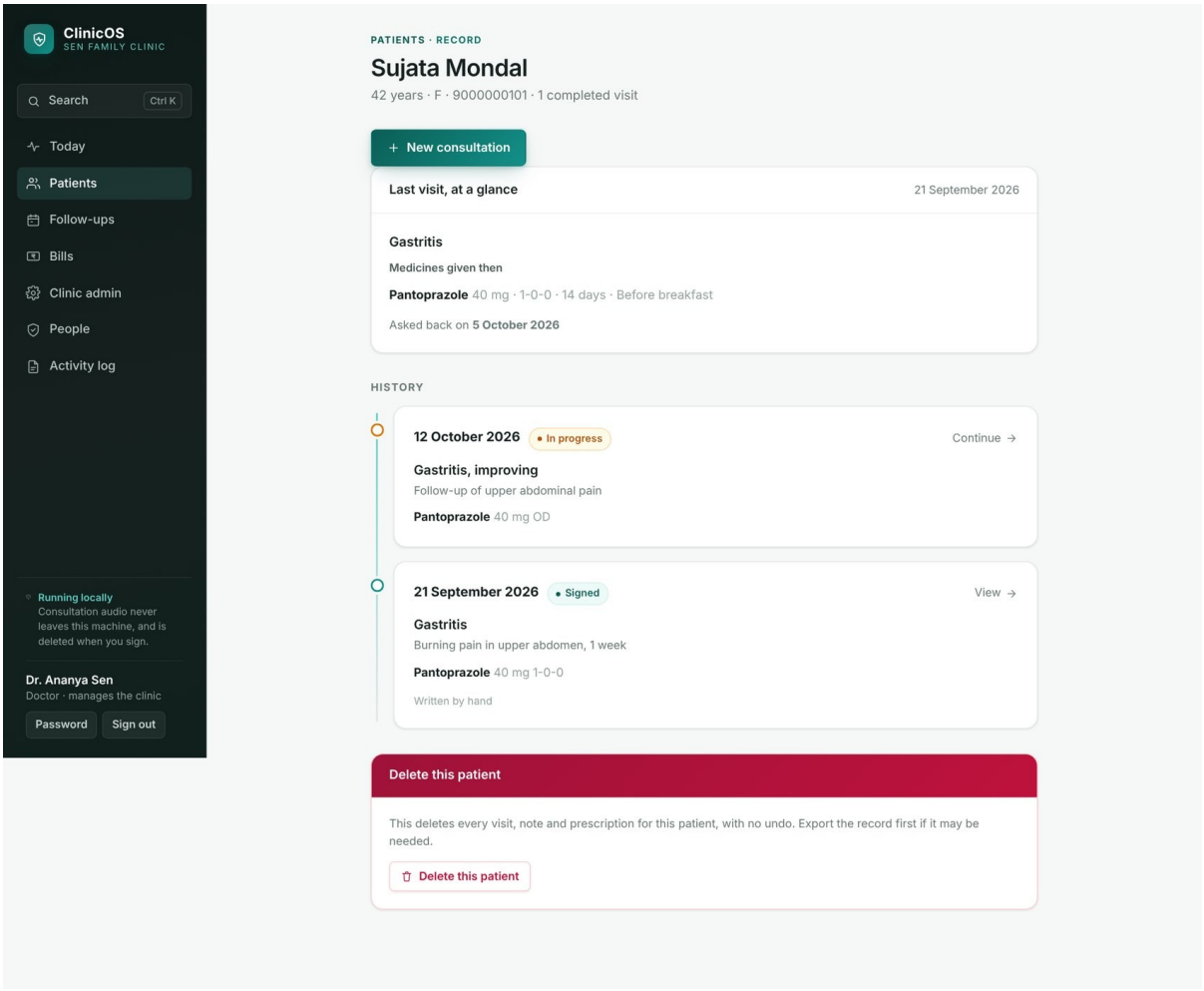


Figure 9: *A patient's record: the last visit at a glance and the full history. Deleting a patient needs their name typed to confirm.*

6.3 The Front Desk

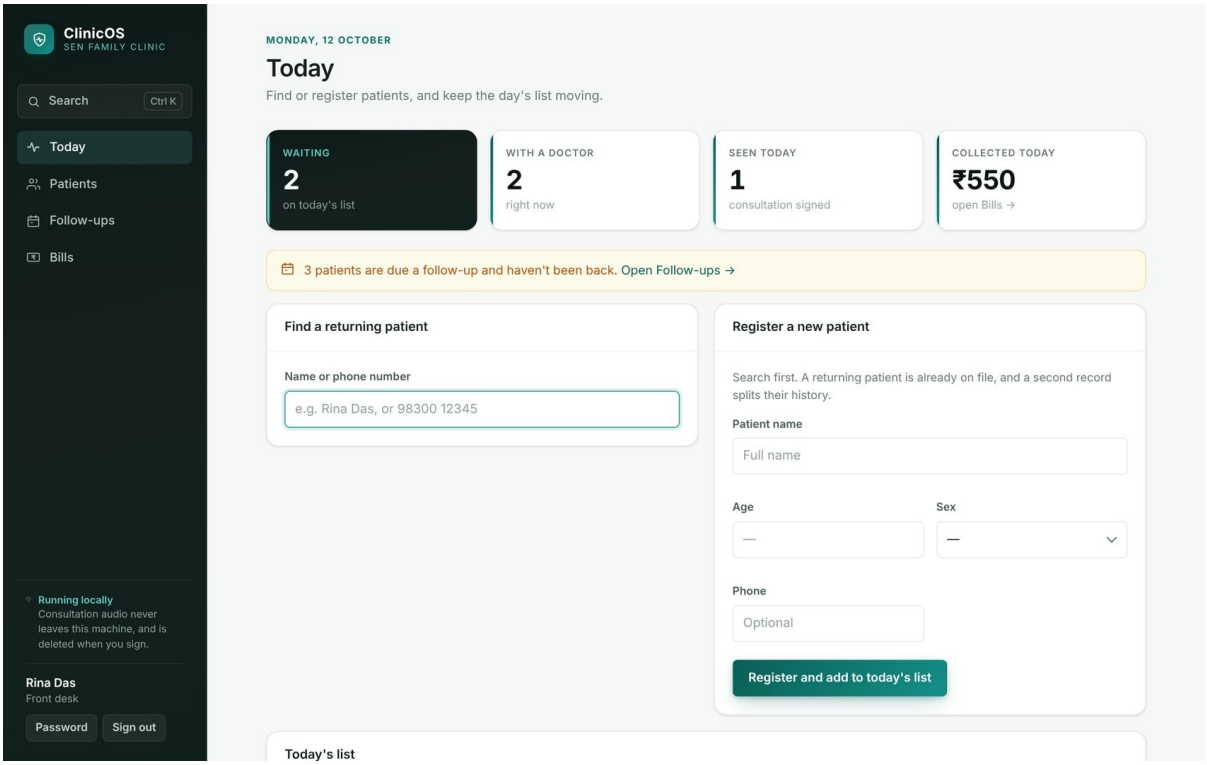


Figure 10: *The front desk's Today: find or register a patient, the day's numbered list, and money collected today. Nothing clinical reaches this screen.*

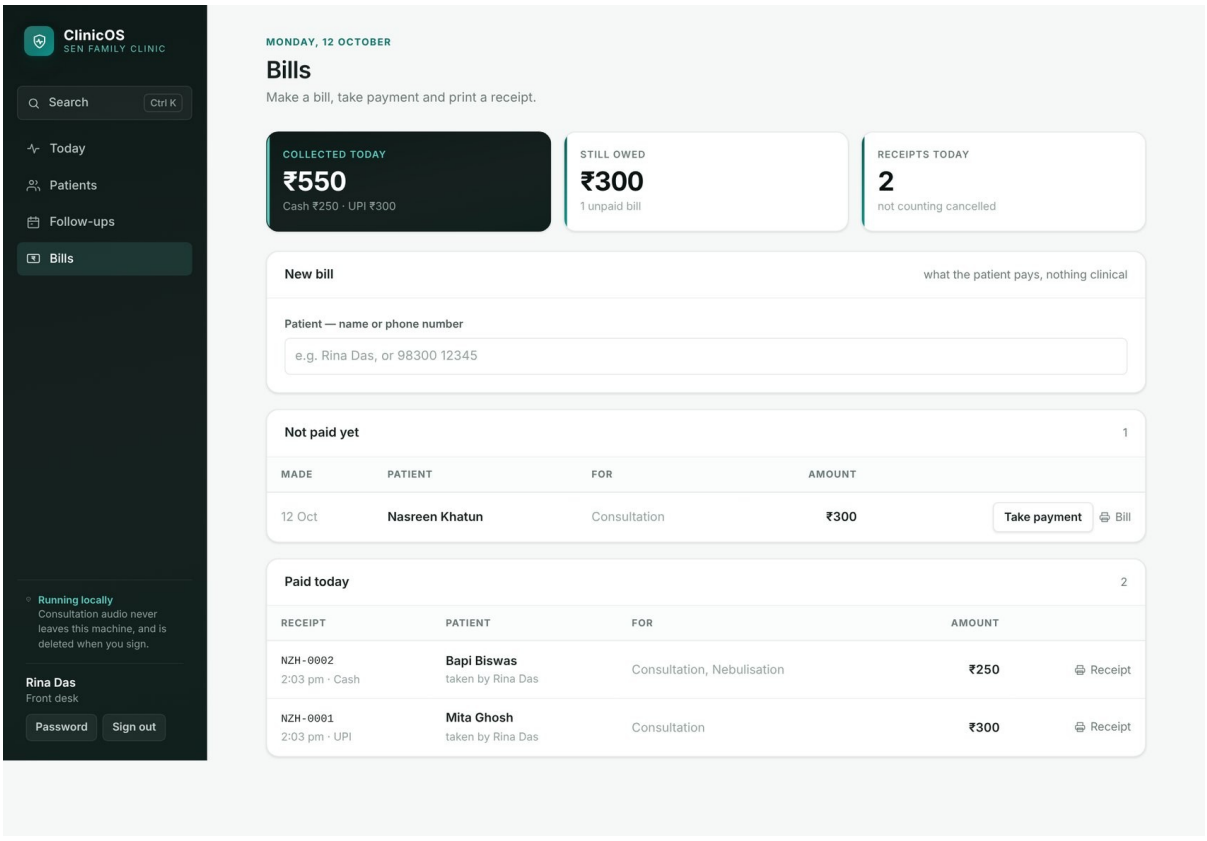


Figure 11: *Bills: collected today by method, what is still owed, and today's receipts.*



Figure 12: *A printed receipt — the clinic, the receipt number, the lines and the total. Nothing clinical is printed on it.*

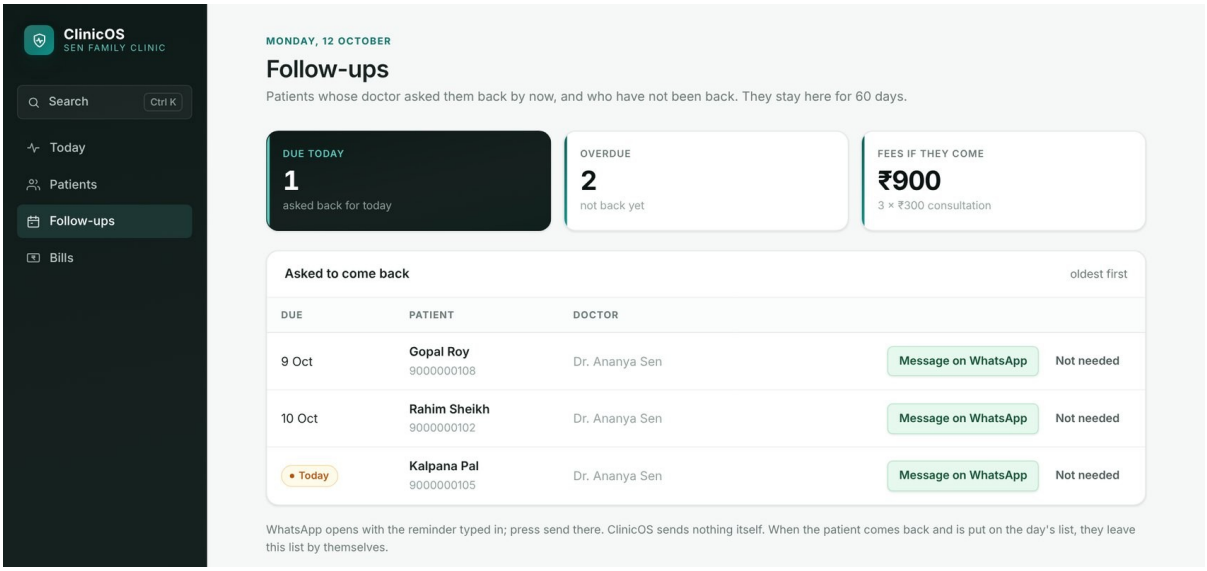


Figure 13: *Follow-ups: patients asked back who have not returned. Message on WhatsApp opens WhatsApp with the reminder typed in; a person presses send.*

6.4 Clinic Admin

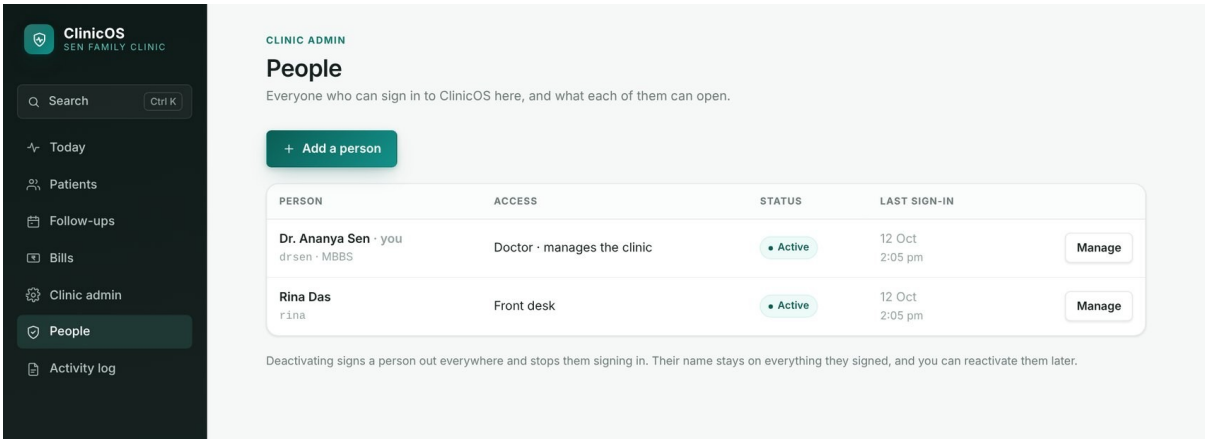


Figure 14: *People: each account and what it can open.*

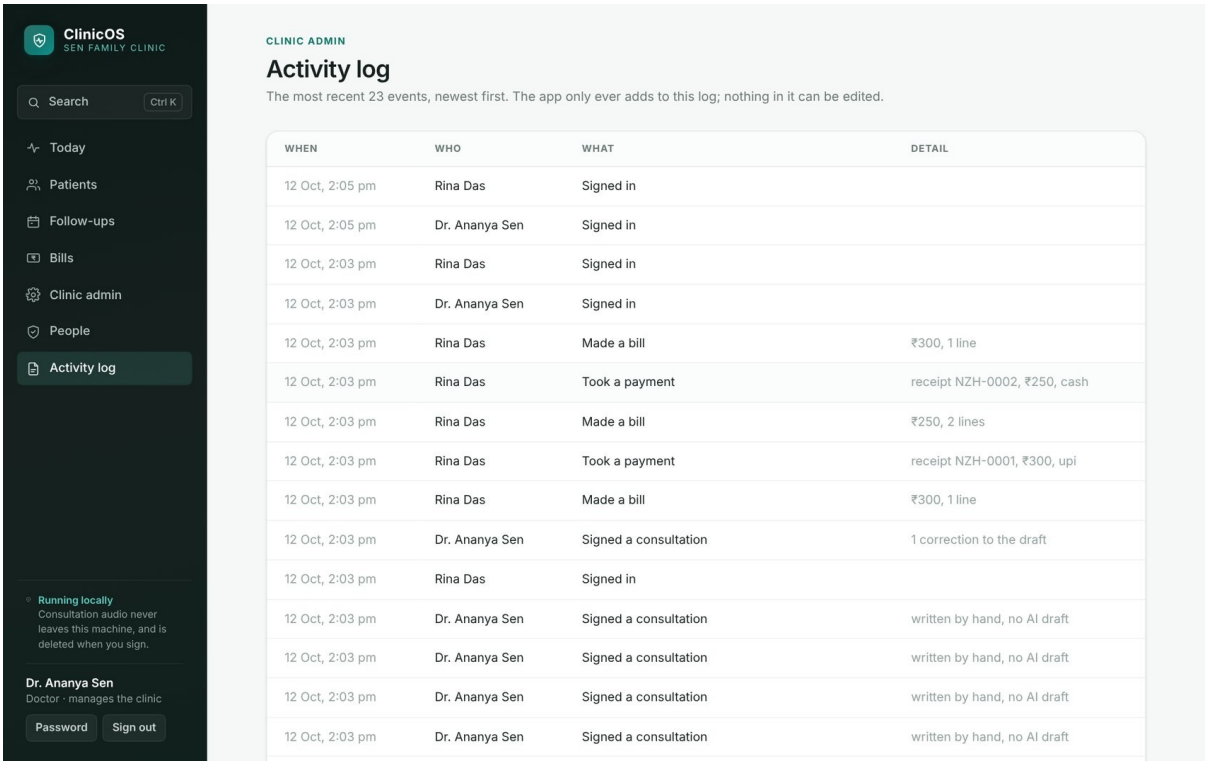


Figure 15: *The activity log records who did what and when. It never contains a patient's name.*

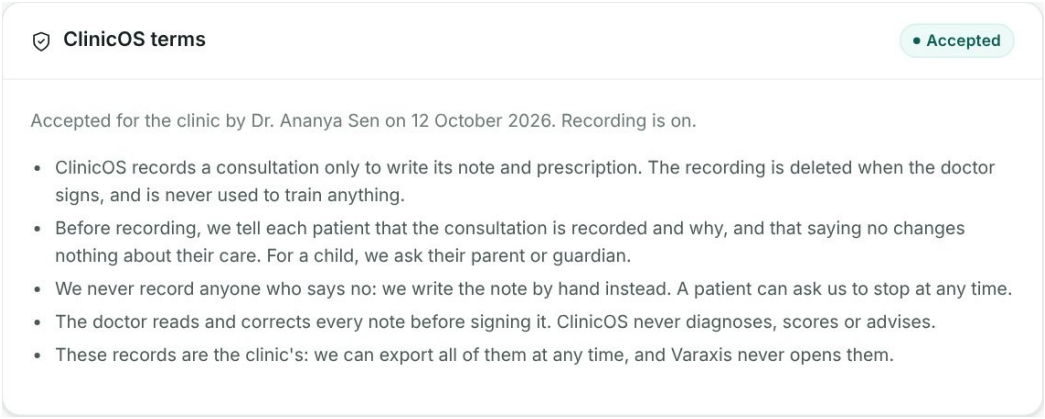


Figure 16: *ClinicOS terms, accepted once by the clinic. Until they are accepted the server refuses to record.*

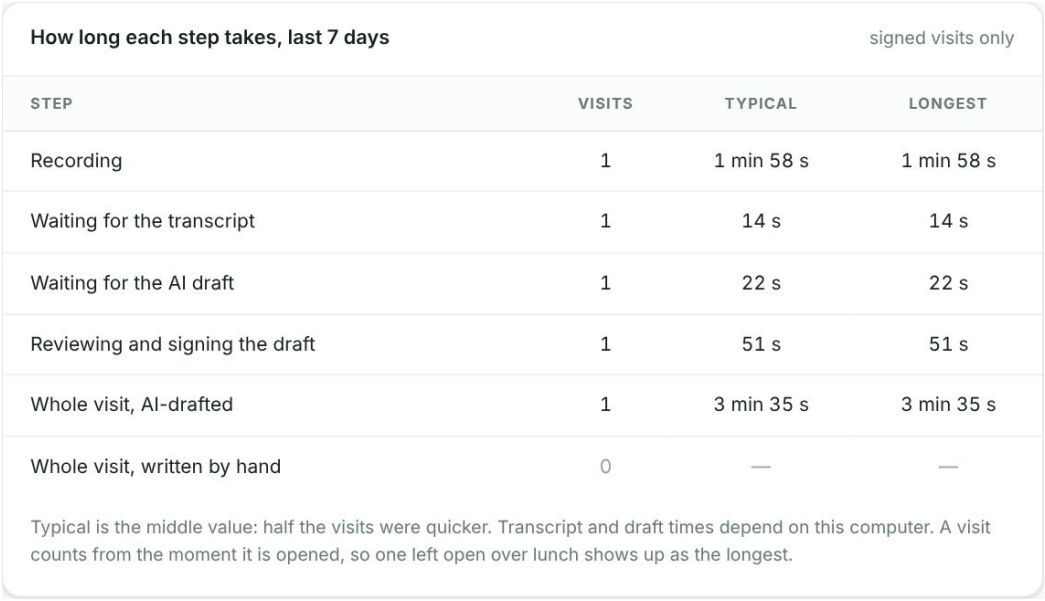


Figure 17: *How long each step of a consultation takes — the middle value and the longest, never an average (sandbox sample data, not a field measurement).*

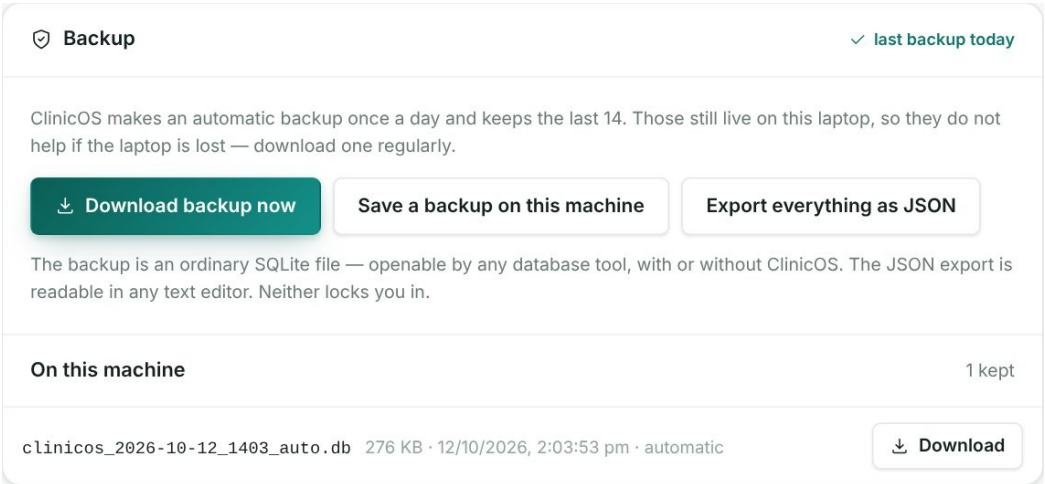


Figure 18: *Backup: a daily automatic snapshot, a one-tap download, and a full export as JSON.*

7. CONCLUSION AND LEARNINGS

Design thinking changed what ClinicOS is. We started by assuming that rural clinics were offline and that an electronic record would fix the paperwork. The Empathize stage broke both assumptions: the real constraint is what a dropped connection costs mid-consultation, and software that makes a doctor type takes time instead of returning it. The Define stage turned that into one problem statement and a point of view centred on the doctor's two minutes. The Ideate stage produced 37 ideas and rejected three of four directions on evidence. The Prototype stage produced working software tested by hundreds of automated checks, and the Implement stage turned it into a product a whole clinic can run — with its ethical limits enforced in code rather than in a policy document.

Our main learnings were:

- **Constraints shape good design.** The legal line between carrying a record and judging it decided what we built and, just as importantly, what we refused to build.
- **Ask about the past, not the future.** Evidence comes from what a doctor did yesterday, not from what they say they would do.
- **Set the pass mark before the test.** Gates written in advance keep a team honest about whether the idea is working.
- **Honest numbers earn trust.** A conservative figure a doctor believes is worth more than an impressive one they doubt.

The most important step is still ahead: the twenty doctor interviews. They will tell us whether the problem we defined is the one doctors actually feel — and the design is built to change if it is not.

8. REFERENCES

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